

Education

Bachelor of Science in Data Science

Iowa State University, Ames, IA

Aug 2023 – Present

Master's of Science in Mechanical Engineering

Iowa State University, Ames, IA

Jan 2025 – Present

EXPERIENCE

Undergraduate Research Assistant

Translational AI Center, Iowa State University

Apr 2024 – Present

- Collaborating with Professor Soumik Sarkar on leveraging LLMs in finance to enhance decision-making processes.
- Gaining insights into AI integration in financial applications, and developing advanced skills in machine learning.

Undergraduate Software Developer

Iowa State University Department of Agricultural and Biosystems Engineering

June 2024 – December 2024

- Worked with Professor Ryan McGhee on developing a versatile chatbot frontend to improve UI and experience.
- Engaged in software development and user interface design.

Undergraduate Research Volunteer

ISU Department of Computer Science

Apr 2024 – May 2024

- Explored the behavior of large language models with knowledge conflicts under Professor Qili's guidance.
- Developed key skills in data research and analysis.

PUBLICATIONS & PRESENTATIONS

Federated In-Context Prompt Selection for Multi-Modal 3D Dental Imaging: A Theoretical Framework with Privacy-Preserving Guarantees

ODIN 2025 (Oral and Dental Image aNalysis) Workshop @ MICCAI 2025 (29th International Conference on Medical Image Computing and Computer Assisted Intervention), Poster

- Developed a novel federated learning framework for multi-modal 2D and 3D dental imaging with privacy-preserving guarantees.
- Key contributions: cross-modal prompt alignment, hierarchical multi-modal optimization, and Byzantine-resilient aggregation with differential privacy.

LLM-based Insights into How Market Orientation and Brand Storytelling Affect Investors' Evaluation of Startup Pitches

REU Summer Symposium 2025, Poster; TrAC Day 2025, Poster & Flash Talk

- Presented research poster at REU Summer Symposium 2025 while serving as a peer mentor to Taj Singh, enhancing both academic and personal growth within the Translational AI Center.
- Delivered a flash talk at TrAC Day 2025, exploring how passion, responsiveness, and storytelling drive investor decisions using GPT-4o and Shark Tank data.

Enhancing Medical AI Safety Through Multi-Agent Evaluation and Iterative Refinement

The 16th ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM-BCB) 2025, Pennsylvania, USA, Poster

- Developed a novel multi-agent refinement framework to enhance the ethical, safety, and regulatory compliance of medical LLM outputs.
- Framework pairs DeepSeek R1 and Med-PaLM as generative models with LLaMA 3.1 and Phi-4 as evaluator agents, iteratively refining responses against the American Medical Association's nine Principles (AMA-9) and a five-tier Safety Risk Assessment (SRA-5).
- Strengthened reliability and alignment of medical LLM outputs, addressing real-world deployment challenges.

LlamaHealth: Assessing LLaMA's Performance in Multi-Class Healthcare Test Result Prediction

ICERIE 2025, Shahjalal University of Science and Technology, Sylhet, Bangladesh, Poster

- Earned Certificate of Participation; reinforced focus on responsible AI for healthcare.
- Investigated LLaMA's multi-class prediction capabilities in healthcare datasets.

Benchmarking Bioemu-1: Establishing a Baseline for Protein Structural Ensemble Prediction

AIChE Annual Meeting 2025, Boston, USA, Poster

- Presented in the Computational Molecular Science and Engineering Forum poster session.

Leveraging Vision Language Models for Specialized Agricultural Tasks

Proceedings of the 2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)

- Explored how vision-language models improve performance on specialized agricultural benchmarks.

The Impact of Market Orientation and Brand Storytelling on Shark Tank Evaluations

Journal of Business & Industrial Marketing (JBIM)

- Analyzed how storytelling and market orientation influence investor evaluation using LLM analysis.

INTERNSHIPS

Undergraduate Summer Research Intern – Translational AI Center

May 2024 - July 2024

HONORS & AWARDS

Dean's High Impact Award for Undergraduate Research (Fall 2025)

- Recognized for outstanding contributions to undergraduate research at Iowa State University.
- Highlighted for innovation and rigor in research area/project.
- Mentored or collaborated with peers to enhance research outcomes.

Dean's List (Fall 2024)

- Achieved distinction for academic excellence across all courses.
- Maintained high GPA while engaging in research and extracurricular activities.

Best-In-Show, George Washington Carver Poster Challenge (2025)

- Awarded top recognition for research poster presentation at the annual Carver Poster Challenge.
- Selected from participants based on clarity, impact, and originality of research.

Borlaug Poster Competition (2025)

- Developed research under TrAC – Translational AI Center exploring how AI and FinTech empower entrepreneurs addressing global food and sustainability challenges.
- Recognized as Undergraduate division 1st place by Iowa State University College of Agriculture and Life Sciences for connecting technology, entrepreneurship, and global food systems.

Bangladesh–Sweden Trust Fund Travel Grant (2025)

- Competitive international grant supporting research travel and presentation opportunities.
- Enabled participation in conference/seminar/workshop and networking with global researchers.

ISU LAS Travel Grant (\$400, AIChE 2025)

- Awarded by the College of Liberal Arts and Sciences, Iowa State University.
- Supported conference presentation at AIChE 2025 and engagement with leading experts.

ACADEMIC SERVICE

Reviewer, LaTeX in Computer Vision Workshop @ CVPR 2025

- Reviewed submissions focused on computer vision applications and methodologies.
- Emphasized promoting diversity and representation in AI research.

Reviewer, Second Workshop on Bangla Language Processing (BLP) @ IJCNLP-AACL 2025

- Reviewed submissions for the workshop co-located with IJCNLP-AACL 2025 in Mumbai, India.
- Contributed to advancing natural language processing research for Bangla language.

Reviewer, LatinX in AI (LXAI) Workshop @ NeurIPS 2025

- Evaluated papers and workshop submissions at prominent AI conference.
- Supported efforts to advance inclusion and emerging research within the AI community.

Ethics Reviewer, NeurIPS 2025 Datasets and Benchmarks Track

- Served as an ethics reviewer for dataset and benchmark papers.
- Ensured compliance with ethical research practices and responsible AI deployment.